

Language-Conditioned Human-in-the-loop Control for A Novel Auxiliary Cargo Handling Method

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Abstract—The loading and unloading of heavy goods in narrow cuboid spaces, such as truck carriages, is an essential yet physically demanding task in logistics and industrial fields. Traditional robotic assistive systems typically rely on rigid heuristic rules or fixed-parameter impedance controllers. These methods struggle to adapt to dynamic task contexts (e.g., handling fragile items or operating in cramped spaces) and require tedious manual tuning, while end-to-end reinforcement learning approaches face challenges in training convergence and safety guarantees. To address these limitations, we propose a novel Large Language Model (LLM)-tuned hierarchical control framework for a custom-built active XYZ suspension platform. The core innovation is a semantic-to-parameter mapping mechanism that translates human natural language instructions into dynamic control parameters (e.g., impedance gains, velocity limits, and virtual safety boundaries) for the low-level execution layer. This architecture allows the robot to achieve “zero-shot” physical adaptation to complex tasks without any domain-specific retraining. Extensive experiments validate that, compared to the optimal traditional impedance baseline, our framework significantly enhances physical human-robot interaction (pHRI). It achieves a 513% improvement in zero-shot success rate for handling liquid/fragile objects, completely eliminates position drift during semantic hovering, and reduces system stabilization time by 60% during false trigger recoveries.

Index Terms—Heavy-load handling, impedance control, efficiency optimization, LLLM-Tuned, Human-machine interaction.

I. INTRODUCTION

NOWADAYS, efficient and safe load handling [1] remains a critical challenge in industrial, medical, and daily assistive contexts, particularly within narrow and unstructured environments such as truck carriages or shipping containers. While prolonged manual handling significantly increases the risk of occupational musculoskeletal disorders, traditional heavy-duty machinery (e.g., forklifts or rigid industrial manipulators) is often rendered unviable due to strict spatial constraints. Although wearable assistive devices and collaborative robots have emerged to bridge this gap, they predominantly rely on fixed-parameter impedance controllers or rigid heuristic state

machines. These conventional approaches lack the cognitive flexibility to dynamically adapt to varying physical contexts—such as handling fragile payloads, responding to sudden spatial limitations, or recovering from sensor anomalies—without tedious manual recalibration. Therefore, developing an assistive handling system that synergistically combines heavy-lifting capacity with human-like semantic adaptability and physical compliance represents an urgent and open challenge.

A. Related Work

Traditional Impedance and Heuristic Control. Conventional robotic assistive systems predominantly utilize PID or Cartesian impedance controllers to manage physical human-robot interaction (pHRI). While these methods provide reliable mathematical stability and computational efficiency, they rely heavily on fixed stiffness, damping, and velocity parameters meticulously pre-tuned by domain experts. Consequently, these rigid models struggle to adapt to dynamic task conditions—such as varying payload weights, fragile cargo, or sudden changes in human intent—without requiring tedious manual recalibration.

Learning-Based Control and Reinforcement Learning. To enable dynamic adaptation and overcome the limitations of fixed heuristics, data-driven approaches—particularly Deep Reinforcement Learning (RL) and Proximal Policy Optimization (PPO)—have been widely explored. Despite their success in simulation, end-to-end RL algorithms suffer from notoriously low sample efficiency. More critically, the inherent “sim-to-real” reality gap and the black-box nature of neural policies make it exceedingly difficult to provide absolute, verifiable safety guarantees, which are non-negotiable in heavy-load human-robot collaborative environments.

LLM-Empowered Robotic Control. Recently, the integration of Large Language Models (LLMs) and Vision-Language-Action (VLA) architectures has emerged as a transformative frontier in robotics, enabling zero-shot reasoning and semantic task planning. However, seamlessly bridging the gap between high-level language reasoning and high-frequency, safe physical execution remains an open challenge. Serving as the exact entry point of this paper, our work proposes a hybrid paradigm: we leverage the cognitive adaptability and semantic context-awareness of LLMs to dynamically orchestrate the parameters of a heuristic low-level controller. This approach achieves

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zero-shot task adaptation while strictly maintaining the mathematical safety boundaries of traditional control.

B. Our Contributions

The main contributions of this paper are summarized as follows:

- **Novel Auxiliary Handling System:** We propose a novel heavy load handling method and develop a custom active XYZ suspension platform integrated with sensorized robotic gloves. Standardized experiments validate its efficacy in significantly reducing human physical exertion and efficiency improvement during heavy-load handling tasks.
- **Systematic Formulation of Traditional Baselines:** We systematically formulate and evaluate various low-level control strategies and select the one with the strongest comprehensive performance as the baseline.
- **LLM-Tuned Hierarchical Control Architecture:** We develop a new human-machine interaction control method with natural language interaction capability. We propose a 'semantic-to-physics' hierarchical control framework with a 'instruction-to-parameter mapping' mechanism. This framework utilizes Large Language Models (LLMs) to dynamically translate natural language instructions into low-level control parameters, enabling zero-shot task adaptation and context-aware safety without requiring any domain-specific retraining.
- **Comprehensive pHRI Evaluation Framework:** We establish a rigorous, objective evaluation framework tailored for physical Human-Robot Interaction (pHRI) in cargo handling, and clarify the definitions of each evaluation indicator. Through quantitative metrics and real-time case studies, we explicitly demonstrate the proposed framework's superior performance.

II. THE NOVEL AUXILIARY CARGO HANDLING METHOD

A. Task Description and Novel Handling Method

The loading and unloading of heavy cargo in narrow, unstructured cuboid spaces (such as truck carriages or shipping containers) poses a significant challenge in modern logistics. Traditional bulky equipment, such as forklifts or rigid industrial manipulators, are often severely restricted by spatial constraints. Consequently, these tasks heavily rely on manual labor, leading to low efficiency and a high risk of occupational musculoskeletal disorders for workers. To address this dilemma, we propose a novel human-robot collaborative cargo handling method. The core concept revolves around a ceiling-mounted active XYZ suspension platform cooperating with a pair of wearable, sensorized robotic gloves via a flexible wire rope.

The operational workflow of this method is highly intuitive: First, the human operator wears the robotic gloves and approaches the cargo to establish a natural handling

posture. Once the cargo is grasped, a mechanical self-locking mechanism on the gloves is engaged to secure the grip. Subsequently, the active Z-axis actuator reels in the suspension rope, bearing the primary gravitational payload of the heavy object. During the transportation phase, the human operator only needs to provide minimal lateral forces to guide the trajectory and adjust the object's orientation, while the active X-Y axes automatically follow the operator's horizontal movement. This hybrid method synergistically combines the kinematic dexterity and cognitive adaptability of a human worker with the heavy-lifting capacity of a robotic hoist, offering an optimal, ergonomic solution for complex logistic environments.

B. Static Force Reduction Analysis

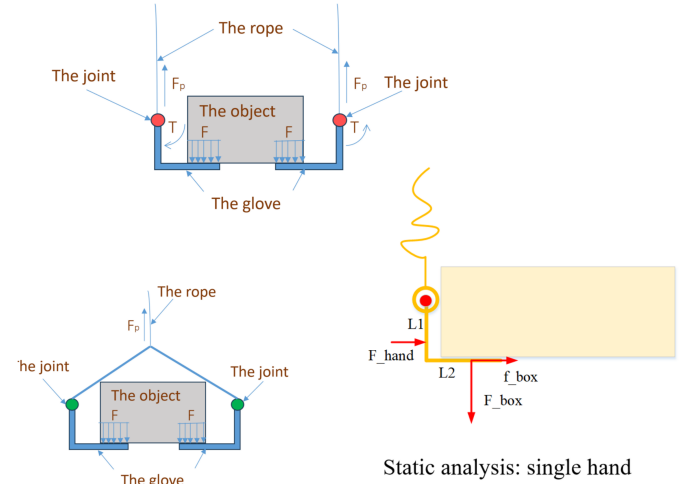


Fig. 1. The static analysis of labor-saving principle.

To quantitatively evaluate the ergonomic benefits of the proposed system, we conduct a static force analysis (see Fig. 1). In traditional manual handling, the operator must exert a continuous vertical lifting force $F_{manual} = W_{cargo} + W_{glove}$ alongside a substantial grip force, rapidly inducing muscle fatigue.

With our active suspension platform, the vertical static equilibrium is restructured as $F_{human_z} + F_{rope} = W_{cargo} + W_{glove}$. Once the mechanical self-locking mechanism of the gloves engages, the need for sustained human grip strength is completely eliminated. Furthermore, the adaptive Z-axis controller actively regulates the rope tension F_{rope} to bear the vast majority of the payload ($F_{rope} \approx W_{cargo} + W_{glove}$). Consequently, the required vertical human exertion is drastically minimized to $F_{human_z} \approx 0$. In this collaborative paradigm, the operator is emancipated from heavy lifting, acting primarily as a kinematic guide by providing only minimal lateral forces (F_{human_x}, F_{human_y}) to adjust the cargo's trajectory.

C. Construction of the platform prototype

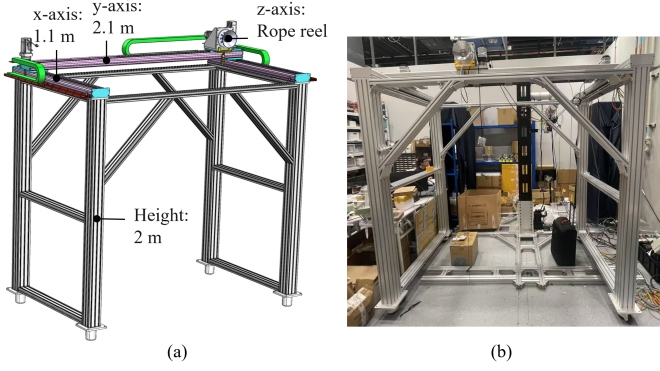


Fig. 2. (a) 3D Model of the active XYZ platform for suspension. (b) The corresponding prototype.

1) Construction of the active XYZ Suspension Platform:

As shown in the Fig. 2, we develop a suspension platform with XYZ three axes. Among them, both the X and Y axes are composed of linear joints, and the Z axis is a rotational joint. The Z-axis is equipped with a rope winding mechanism at its end, which can lift or release the rope. The purpose of this platform is to cooperate with our developed robotic gloves to achieve the auxiliary handling function. The specific steps are as follows: (1) Human workers take on our robotic gloves and move to the target position, then pose for handing; (2) Gently lift the target object until the sensor can detect the signal; (3) After receiving this signal, the controller controls the Z-axis to rotate forward to pull up the rope. It can lift the two gloves that have already been self-locked, thereby achieving the lifting of the target object. At this point, the gravity of the target object is mainly borne by the rope connected to the Z-axis (on the winding wheel), and the human workers only need to provide a small amount of support and lateral force to maintain balance, thereby successfully achieving the effect of auxiliary handling.

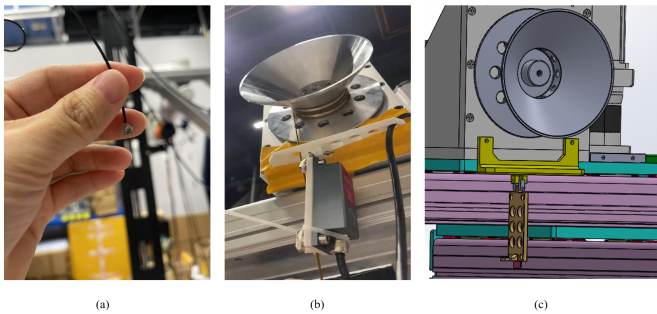


Fig. 3. Sensors equipment of the XYZ platform. (a) The uniaxial force sensor (for Z-axis). (b) The horizontal inclinometer (for X-Y axes control). (c) 3D model of the universal joint (for connecting the inclinometer).

2) The Sensors Equipment Schemes:

As shown in Fig. 3(a-b), the platform is equipped with a miniature uniaxial force sensor and a horizontal inclinometer. The force sensor, mounted between the wearable and mechanical gloves, measures the human lifting force to dynamically

regulate the active Z-axis tension. The inclinometer, attached to the Y-axis via a custom 2-DOF passive rotating mechanism (Fig. 3(c)), detects the horizontal deflection of the suspension rope. This deflection data drives the X and Y axis motors via a proportional-derivative (PD) controller [2], [3], enabling the platform to actively track the operator's horizontal movements. This active 3D compliance significantly expands the operational workspace and minimizes physical exertion compared to traditional passive single-axis hoists. Furthermore, a Voice/Audio Input Module is integrated into the system to facilitate higher-level cognitive interaction. Through a wearable microphone or a mobile application, operators can issue natural language instructions (e.g., "This is heavy, pull it slowly") to the system. This allows the framework to dynamically adapt its low-level control parameters based on real-time semantic contexts.

III. THE PROPOSED CONTROL LOGICS

To achieve seamless human-robot collaboration during the cargo handling process, the control strategy of the active XYZ suspension platform must accurately interpret human intent and provide corresponding assistive forces. In this section, we first formulate the evolution of our low-level traditional control baselines, and then introduce the proposed Large Language Model (LLM)-tuned hierarchical architecture.

A. Low-Level Baseline Strategies

Based on the sensory inputs from the rope tension F_{rope} and human fingertip pressure F_{touch} , the fundamental goal of the Z-axis controller is to generate a target motor velocity V_{goal} or torque. In the initial development phase, the baseline system relied on rigid state machines and manual parameter tuning to dictate motor behaviors. We abstract our previous heuristic and classic control implementations into the following categories:

1) Velocity-based Heuristic State Machine (VHSM):

Operating in the speed mode, the system transitions between three rigid states (Ascending, Hovering, and Descending) triggered by hard-coded pressure thresholds. The target lifting velocity V_{goal} is computed as:

$$V_{goal}(t) = \begin{cases} -K_v \cdot F_{touch}(t), & \text{if } F_{touch}(t) > \tau_{up} \\ 0, & \text{if } F_{touch}(t) \leq \tau_{down} \\ V_{descend}, & \text{otherwise} \end{cases} \quad (1)$$

where τ_{up} and τ_{down} define the ascending and hovering activation bounds, K_v is the proportional velocity gain, and $V_{descend}$ is a constant downward release speed.

2) Fixed-Target Torque-based PID Tracking (FTTPT):

Operating in the motor's torque mode, the required assistive torque τ_{goal} is generated proportionally to the user's fingertip pressure F_{touch} . To counteract the static gravity of the system itself, a manually calibrated balance value

$F_{balance}$ is subtracted. An upper saturation limit τ_{max} is applied to ensure hardware safety:

$$\tau_{goal}(t) = \min \left(\max (K_{\tau} \cdot (F_{touch}(t) - F_{balance}), 0), \tau_{max} \right) \quad (2)$$

where K_{τ} is the fixed force-to-torque gain. *Limitation:* While computationally efficient, these heuristic methods require continuous manual adjustment of $F_{balance}$ or threshold parameters for different loads [4], [5]. A fixed parameter set tailored for a 10kg box inevitably causes severe oscillations or sluggish responses when handling a 40kg load.

3) *Adaptive Proportional Control with Dynamic Weight Estimation (APC):* To overcome the limitations of fixed-weight tracking, we developed an adaptive measurement and control subroutine. The system's behavior is governed by a heuristic three-state transition logic based on the human fingertip pressure F_{touch} :

- **Measurement State (Blind Lifting):** Triggered when F_{touch} exceeds the threshold τ_{up} while the system is currently unloaded. The platform actively reels in the rope at a constant initialization speed. Once the rope tension F_{rope} surpasses a safety limit (indicating the object is fully suspended), the system enters a brief sampling window. The estimated cargo weight $\hat{W}$ is derived using a discrete moving average of N samples to filter out transient dynamic jitter:

$$\hat{W} = \alpha \left(\frac{1}{N} \sum_{i=1}^N F_{rope}(t_i) \right) + \beta \left(\frac{1}{N} \sum_{i=1}^N F_{touch}(t_i) \right) \quad (3)$$

where α and β are fusion coefficients based on static load distribution analysis. $\hat{W}$ then serves as the adaptive equilibrium setpoint for the subsequent handling task.

- **Load State (Adaptive Carrying):** Once $\hat{W}$ is successfully estimated and the operator maintains a stable grip ($\tau_{down} \leq F_{touch} \leq \tau_{up}$), the system transitions into the load-carrying state. The tracking error is calculated using a multi-point moving average of the rope tension $\bar{F}_{rope}(t)$ against the estimated weight $\hat{W}$. To suppress micro-vibrations induced by human pulse or slight shaking, a deadband ϵ is introduced:

$$e(t) = \begin{cases} 0, & \text{if } |\bar{F}_{rope}(t) - \hat{W}| < \epsilon \\ \bar{F}_{rope}(t) - \hat{W}, & \text{otherwise} \end{cases} \quad (4)$$

The target velocity command is generated proportionally: $V_{cmd}(t) = K_p \cdot e(t)$. To ensure execution smoothness and the command remains within physical limits, we first define the constrained velocity increment:

$$\Delta V_{lim} = \text{clip}(\Delta V_{cmd}, -\Delta_{max}, \Delta_{max}) \quad (5)$$

Subsequently, the target velocity is updated as follows:

$$V_{goal}(t) = \text{clip}(V_{goal}(t-1) + \Delta V_{lim}, -V_{max}, V_{max}) \quad (6)$$

where Δ_{max} represents the maximum allowable acceleration step.

- **Loosen State (Release):** When the worker intends to place the cargo and releases their grip ($F_{touch} < \tau_{down}$), the system immediately flushes the estimated weight flag ($\hat{W} \leftarrow 0$) and transitions to a slack-tracking mode. The target velocity rapidly decreases to prevent the active Z-axis from dragging the human worker upwards, ensuring a safe detachment from the cargo.

4) *Fixed-Parameter Impedance Control Based on Pressure (FPIC):* To improve compliance during the human-robot physical interaction, an impedance control law is implemented on top of the dynamic weight estimation. The interactive force exerted by the human, F_{human} , is defined as the difference between the estimated weight and the current rope tension: $F_{human}(t) = \hat{W} - F_{rope}(t)$. We model the desired interaction dynamics as a mass-damper system:

$$M_d \dot{V}_{goal}(t) + B_d V_{goal}(t) = F_{human}(t) \quad (7)$$

where M_d is the desired virtual mass and B_d is the desired damping coefficient. The discrete time update law for the actuator velocity is strictly constrained by hardware acceleration limits A_{max} :

$$a^*(t) = \text{clip} \left(\frac{F_{human}(t) - B_d V_{goal}(t-1)}{M_d}, -A_{max}, A_{max} \right) \quad (8)$$

$$V_{goal}(t) = V_{goal}(t-1) + a^*(t) \cdot \Delta t \quad (9)$$

Limitation: Although this adaptive impedance controller (acting as our *Best Traditional* baseline) provides smoother handling than purely proportional methods, the impedance parameters ($M_d = 15, B_d = 0.1$) remain static. It still cannot adapt to sudden semantic context changes, such as switching from "fast moving" to "fragile placement." This limitation necessitates the introduction of a high-level cognitive planner.

B. Hierarchical LLM-Based Control Architecture

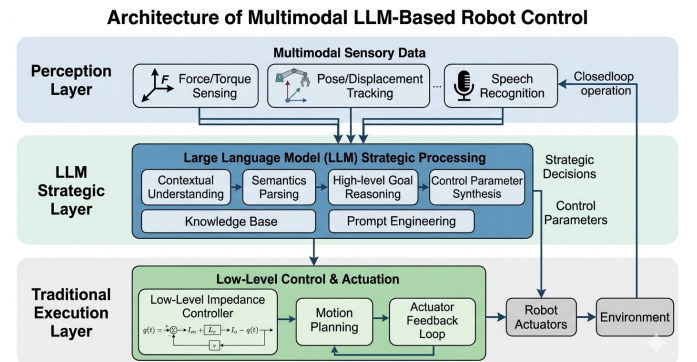


Fig. 4. Architecture of the multimodal LLM-based robot control framework, illustrating the data flow from human perception to semantic parameter generation, and finally to low-level execution.

To overcome the inherent inflexibility of fixed-parameter baselines, we propose a novel multimodal hierarchical control architecture that seamlessly bridges high-level cognitive reasoning with low-level deterministic execution. As illustrated in Fig. 4, the proposed framework is structurally partitioned into three interconnected layers:

- **Perception Layer:** This foundational layer is responsible for multimodal data acquisition. It continuously captures high-frequency physical states—such as rope tension (F_{rope}), human fingertip pressure (F_{touch}), and Z-axis spatial displacement (Z_{cur})—alongside asynchronous human semantic intents collected via the audio input module.
- **LLM Strategic Layer:** Serving as the cognitive core, this layer evaluates the operator’s natural language instructions against the real-time physical context. Instead of generating direct, uninterpretable motor torques, the LLM is tasked with outputting a strictly formatted JSON configuration packet ($\mathcal{P}_{llm}$). To ensure output stability and reproducibility, the LLM operates under a rigorously designed system prompt. A condensed version of our core prompt is formulated as follows: “*You are the cognitive brain of an industrial assistive hoist. Based on the user’s natural language command (U_t) and the current kinematic/dynamic states (S_t), infer the task context. Adjust the impedance gains (K_{llm}, ϵ_{llm}), kinematic limits (ΔV_{max}^{llm}), or establish virtual safety boundaries (Z_{max}^{llm}). Output your parameter modifications strictly in valid JSON format.*”
- **Traditional Execution Layer:** Operating safely at the base, this layer incorporates the robust adaptive impedance controller established in Section III-A. It continuously ingests the JSON parameter packets from the LLM and instantly applies the updated physical constraints to the high-frequency motor control loop.

By deliberately isolating the non-deterministic semantic reasoning from the high-frequency actuator control loop, this hierarchical architecture endows the system with “zero-shot” cognitive adaptability while strictly preserving the mathematical stability and absolute safety boundaries inherent to classical control theory.

C. Semantic-to-Parameter Mapping

The core innovation of our hierarchical control architecture lies in the semantic-to-parameter mapping mechanism. Unlike traditional robotic systems where control parameters (e.g., proportional gains, deadbands, and velocity limits) are hard-coded or manually tuned by engineers, our system utilizes the LLM as a dynamic parameter generator.

The LLM processes the user’s natural language instruction U_t and the current system state S_t , outputting a structured JSON configuration packet $\mathcal{P}_{llm}$. This packet contains specific parameter modifiers that are directly injected into the low-level execution layer. We categorize

the human-robot semantic interaction into five distinct functional modes:

1) *Physical Property Presets (Pre-compensation):* Human workers can provide a priori knowledge about the cargo before physical contact. The LLM translates these property descriptions into feedforward compensations to eliminate sensor lag and prevent sudden jerks. For instance, if the user states, “*This box contains liquid, keep it extremely steady,*” the LLM outputs a constrained acceleration limit ΔV_{max}^{llm} and an initial tension bias V_{ff}^{llm} . The velocity update law is thus modified as:

$$V_{goal}(t) = \text{clip}\left(V_{goal}(t-1) + \Delta v, -\Delta V_{max}^{llm}, \Delta V_{max}^{llm}\right) + V_{ff}^{llm} \quad (10)$$

where $\Delta v = K_{llm} \times e(t)$ is the calculated velocity increment. This ensures zero-spill handling without requiring complex dynamic modeling of the liquid.

2) *Task Mode Switching:* Verbal instructions can forcefully override the low-level state machine. When the user says, “*Hold it right there*” or “*I need to stack this continuously,*” the LLM outputs a state-switching flag $\mathcal{M}_{llm} \in \{\text{Hover, Follow, Rapid_Measure}\}$. For the hovering mode, the control law explicitly bypasses sensor-driven inputs:

$$\text{If } \mathcal{M}_{llm} == \text{Hover} \implies V_{goal}(t) \equiv 0, \quad \forall e(t) \quad (11)$$

This semantic override inherently eliminates position drift and unintended actuator work caused by involuntary muscle tremors when the worker is adjusting their posture, thereby guaranteeing absolute spatial stability during the hovering phase.

3) *Compliance and Sensitivity Regulation:* This category directly adjusts the “feel” of the robot, equivalent to tuning the impedance/admittance characteristics. In the original heuristic control code, the proportional gain and deadband were rigidly fixed (e.g., $V_{goal} = 15 \times err$). Under the proposed framework, if the user instructs, “*Pull it faster*” or “*Be gentle,*” the LLM modulates the proportional gain K_{llm} and the deadband threshold ϵ_{llm} . The active tracking error $e^*(t)$ is recalculated as:

$$e^*(t) = \begin{cases} 0, & \text{if } |e(t)| \leq \epsilon_{llm} \\ e(t) - \text{sgn}(e(t))\epsilon_{llm}, & \text{if } |e(t)| > \epsilon_{llm} \end{cases} \quad (12)$$

The target velocity is then generated by the dynamically assigned gain:

$$V_{goal}(t) = K_{llm} \times e^*(t) \quad (13)$$

A smaller ϵ_{llm} and higher K_{llm} yield an aggressive, highly sensitive following response, whereas opposite configurations provide a damped, compliant interaction.

4) *Safety and Boundary Intervention:* In narrow cuboid spaces (e.g., truck carriages), spatial awareness is critical. Users can issue spatial constraints such as, “*The space is cramped here, restrict the lifting height.*” The LLM translates this into dynamic spatial envelopes $\{Z_{min}^{llm}, Z_{max}^{llm}\}$.

The execution layer applies a strict barrier function to the output velocity:

$$V_{goal}(t) = \begin{cases} \min(0, V_{goal}(t)), & \text{if } Z_{cur} \geq Z_{max}^{llm} \\ \max(0, V_{goal}(t)), & \text{if } Z_{cur} \leq Z_{min}^{llm} \\ V_{goal}(t), & \text{otherwise} \end{cases} \quad (14)$$

This actively prevents collisions before the hardware limit switches are triggered.

5) *Error Correction and State Reset*: Force and pressure sensors are susceptible to accidental touches. A worker might brush against the sensor and trigger a false lifting sequence. With a simple verbal command, “Wait, that was a mistake,” the LLM triggers a reset sequence $R_{llm} = 1$. The controller flushes the moving average buffers and resets the weight estimation logic:

$$\text{If } R_{llm} == 1 \implies \begin{cases} \text{flag_weight} \leftarrow 0 \\ \bar{F}_{touch} \leftarrow 0 \\ V_{goal}(t) \leftarrow 0 \end{cases} \quad (15)$$

This language-conditioned error recovery effectively limits the peak unintended displacement and drastically reduces the system stabilization time compared to manual emergency stops. It provides a software-level safety fallback against sensor anomalies.

Algorithm 1 Asynchronous LLM Semantic Parsing Thread

Require: Voice instruction U_t , Real-time sensor states S_t

Ensure: Parameter configuration dictionary $\mathcal{P}_{llm}$

```

1: Initialize  $\mathcal{P}_{llm} \leftarrow \{K_{llm} = 15, \epsilon_{llm} = 100, \Delta V_{max}^{llm} = 30, \dots\}$ 
2: while True do ▷ Runs asynchronously at low frequency
3:   if Voice_Command_Detected( $U_t$ ) then
4:      $S_t \leftarrow \text{Get\_Current\_States}()$ 
5:     JSON_Response  $\leftarrow \text{Query\_LLM}(U_t, S_t)$ 
6:     Update Global  $\mathcal{P}_{llm} \leftarrow \text{Parse\_to\_Dict}(\text{JSON\_Response})$ 
7:   end if
8: end while
```

IV. EXPERIMENTAL EVALUATIONS

To comprehensively validate the proposed system, our experiments are structured progressively: verifying the hardware’s physical assistance capabilities (Sec. IV-A), establishing the optimal traditional control baseline (Sec. IV-B), quantifying the objective improvements brought by the LLM (Sec. IV-C), and explicitly demonstrating the semantic adaptation mechanisms through detailed case studies (Sec. IV-D).

A. Overall Physical Assistance

1) *Whole Handling Process*: The primary objective of the active XYZ suspension platform is to reduce human physical exertion. The basic handling process using our

Algorithm 2 High-Frequency Adaptive Execution Loop

Require: Real-time sensor states $S_t = \{F_{rope}, F_{touch}, Z_{cur}\}$, Global $\mathcal{P}_{llm}$

Ensure: Actuator target velocity V_{goal}

```

1: while True do
2:   time.sleep( $dt$ )
3:    $F_{rope}, F_{touch}, Z_{cur} \leftarrow \text{Read\_Sensors}()$ 
4:   % 1. Error Correction & Reset
5:   if  $\mathcal{P}_{llm}.R_{llm} == 1$  then
6:     flag_weight  $\leftarrow 0$ , Weight  $\leftarrow 0$ 
7:      $\mathcal{P}_{llm}.R_{llm} \leftarrow 0$  ▷ Clear reset flag
8:   end if
9:   % 2. Task Mode Switching
10:  if  $\mathcal{P}_{llm}.\mathcal{M}_{llm} == \text{Hover}$  then
11:     $V_{goal} \leftarrow 0$  ▷ Force zero velocity
12:  else
13:    % 3. Compliance & Sensitivity Regulation
14:     $e(t) \leftarrow F_{rope} - \text{Weight}$ 
15:    if  $|e(t)| \leq \mathcal{P}_{llm}.\epsilon_{llm}$  then
16:       $e^*(t) \leftarrow 0$  ▷ Dynamic semantic deadband
17:    else
18:       $e^*(t) \leftarrow e(t) - \text{sgn}(e(t)) \cdot \mathcal{P}_{llm}.\epsilon_{llm}$ 
19:    end if
20:    % 4. Physical Property Presets (Feedforward)
21:     $V_{target} \leftarrow \mathcal{P}_{llm}.K_{llm} \times e^*(t) + \mathcal{P}_{llm}.V_{ff}^{llm}$ 
22:    % Apply Adaptive Acceleration Limits
23:     $\Delta v \leftarrow V_{target} - V_{goal}(t-1)$ 
24:     $\Delta v \leftarrow \text{clip}(\Delta v, -\mathcal{P}_{llm}.\Delta V_{max}^{llm}, \mathcal{P}_{llm}.\Delta V_{max}^{llm})$ 
25:     $V_{goal} \leftarrow V_{goal}(t-1) + \Delta v$ 
26:  end if
27:  % 5. Safety Boundary Intervention
28:  if  $Z_{cur} \geq \mathcal{P}_{llm}.Z_{max}^{llm}$  then
29:     $V_{goal} \leftarrow \min(0, V_{goal})$  ▷ Prevent upward movement
30:  else if  $Z_{cur} \leq \mathcal{P}_{llm}.Z_{min}^{llm}$  then
31:     $V_{goal} \leftarrow \max(0, V_{goal})$  ▷ Prevent downward movement
32:  end if
33:  myXYZ.AxisMode_Jog(3, 30,  $V_{goal}$ ) ▷ Execute control command
34:   $V_{goal}(t-1) \leftarrow V_{goal}$ 
35: end while
```

auxiliary system consists of the following phases: putting on gloves, carrying gloves to the target position and posing for handling, gently lifting the target object until the force sensor on the right hand gets a signal (the system will enter a smooth force-controlled lifting mode), following the automatically pulled rope to lift the target object, moving the object horizontally to the target area (the XYZ platform can automatically follow the worker’s movement based on the signal from the tilt sensor), lowering the object, moving the gloves to the two outer sides and releasing the object, end this handling and proceed to the next handling. We visually demonstrate this process using

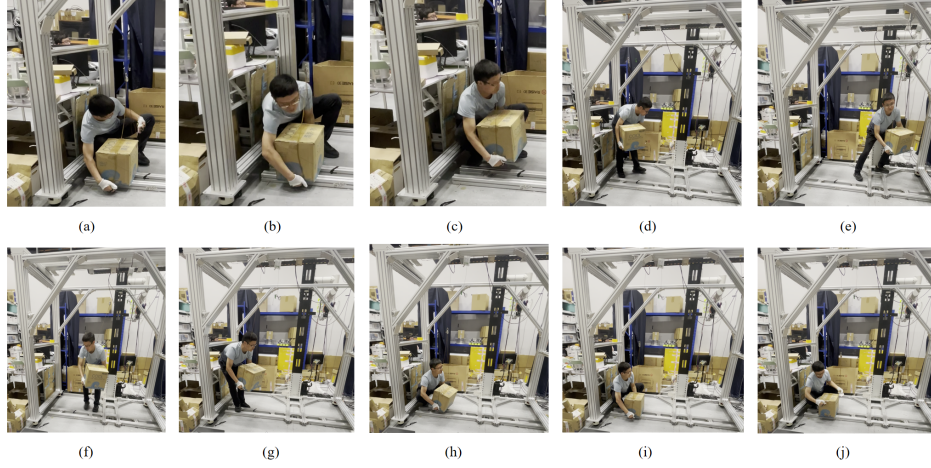


Fig. 5. The whole cargo handling process. (a) Put on, (b) Posing, (c) Lift, (d) Pulled, (e-f) Move, (g-h) Lower, (i) Release, (j) End.

a 20kg carton box in Fig. 5.

2) *Basic Performance Improvement Comparison*: To evaluate the actual improvement in handling efficiency and force reduction, we designed standardized tasks involving two-handed handling (carton boxes) and one-handed handling (suitcases). The specific process is to transport target objects weighing 10kg, 20kg, and 40kg back and forth between the starting and ending areas. Five random participants transported objects weighing 5kg, 10kg, and 20kg back and forth 20 times. We record the required total handling time and the support force required from human hands when using and not using our system, respectively. We also record the overall force saving during the transportation process (represented by $F * t$). As illustrated in Table I, the integration of the robotic system significantly reduces the human support force required and mitigates heart rate elevation across all loads compared to purely manual operation, with negligible compromise to the total task time.

B. Low-Level Control Baselines Evaluation

Before introducing semantic LLM tuning, it is crucial to establish the most robust traditional controller as our comparative baseline. We established an evaluation framework for interface control, focusing on response delay, intent-action match rate, and force jitter during cyclical lifting tasks.

As shown in Table II, purely heuristic methods (e.g., Rope/Hand S) suffer from severe force jitter when operating with mismatched fixed parameters. The *Fixed Impedance Control* with dynamic weighing achieved the best traditional tracking performance with minimal jitter (2.5-4.1N) and robust intent-matching. Consequently, it is selected as the *Best Traditional Baseline* against which the LLM-tuned method is compared.

C. Quantitative Performance of LLM-Tuned Control

Although the fixed impedance baseline ensures fundamental stability, it lacks context awareness. Table III sys-

tematically quantifies the physical performance improvements generated by incorporating the LLM semantic-to-parameter mapping. Evaluated across the five operational modes defined in Sec. III-D, the LLM-tuned architecture achieves a remarkable 513% enhancement in zero-shot adaptation for complex cargo, completely eliminates position drift during semantic hovering, and reduces system stabilization time by 60% during error recovery events.

D. Case Studies of Semantic Adaptation

To explicitly demonstrate the real-time physical performance improvements brought by the LLM tuning, we conducted four objective hardware experiments. We compared the high-frequency kinematic and dynamic states logged during operations under both the traditional *Fixed Impedance* mode and our *LLM-Tuned* framework.

Case 1: Zero-shot Adaptation to Liquid/Fragile Objects (Mode 1 & 3): We tasked the system with lifting an open container of liquid placed on top of a standard carton. Under the traditional fixed impedance control, the instantaneous force application easily saturated the acceleration limits, causing a sharp target velocity spike that resulted in fluid spillage. In contrast, after the operator provided the semantic prompt “*There is water inside, lift it smoothly,*” the LLM proactively reduced the acceleration limit ΔV_{max}^{llm} and increased the impedance damping B_d . The modified velocity curve exhibited a heavily damped, ramp-up trajectory, guaranteeing a spill-free lifting process without any prior domain training.

Case 2: Intent Matching for Hovering (Mode 2): We evaluated the system’s robustness against human fatigue during a mandatory 5-second mid-air hovering task. To simulate operator fatigue, discontinuous, small-amplitude force perturbations were applied to the fingertip pressure sensors. The traditional controller misidentified these tremors as continuous lifting intents, causing the platform to drift vertically. Conversely, upon hearing “*Hold the height,*” the LLM executed a hard state override ($\mathcal{M}_{llm} = \text{Hover}$), forcing the motor velocity to absolute

TABLE I
COMPARISON OF MAIN PERFORMANCE INDICATORS OF VARIOUS CONTROL METHODS

Control Method	Sensors Config	Heart Rate Δ (bpm)			Force Reduction (N)			Task Time (s)		
		5kg	10kg	20 kg	5kg	10kg	20 kg	5kg	10kg	20 kg
Manual Operation	None	+18 $\pm$ 3	+25 $\pm$ 4	+32 $\pm$ 5	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	45 $\pm$ 5	52 $\pm$ 6	58 $\pm$ 7
Velocity-based State Machine	Pressure	+18 $\pm$ 3	+25 $\pm$ 4	+32 $\pm$ 5	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	45 $\pm$ 5	52 $\pm$ 6	58 $\pm$ 7
Fixed-Target PID Tracking	Force	+15 $\pm$ 3	+22 $\pm$ 4	+28 $\pm$ 5	15 $\pm$ 2	25 $\pm$ 3	35 $\pm$ 4	42 $\pm$ 4	48 $\pm$ 5	55 $\pm$ 6
Proportional Control (Adaptive Meas.)	F+P	+8 $\pm$ 3	+14 $\pm$ 4	+21 $\pm$ 5	22 $\pm$ 2	35 $\pm$ 3	48 $\pm$ 4	35 $\pm$ 3	42 $\pm$ 4	49 $\pm$ 5
Hand S (PID)	Pressure	+15 $\pm$ 3	+22 $\pm$ 4	+28 $\pm$ 5	15 $\pm$ 2	25 $\pm$ 3	35 $\pm$ 4	42 $\pm$ 4	48 $\pm$ 5	55 $\pm$ 6
Hand S (Fixed Impedance)	Pressure	+15 $\pm$ 3	+22 $\pm$ 4	+28 $\pm$ 5	15 $\pm$ 2	25 $\pm$ 3	35 $\pm$ 4	42 $\pm$ 4	48 $\pm$ 5	55 $\pm$ 6

¹ All values represent mean $\pm$ standard deviation (n=10 trials per condition).

² Heart Rate Δ : Change from resting baseline (62 bpm) during operation.

³ Force Reduction: Saved human exertion compared to manual lifting baseline.

⁵ Testing conditions: Standardized box lifting (0.5m height, 10 repetitions).

³ Handling Time: Total time required to complete standardized lifting task.

⁴ Sensor Config abbreviations: Force=Force sensor, Pressure=Pressure sensor.

TABLE II
OBJECTIVE COMPARISON (CONTINUED): ADVANCED PERFORMANCE METRICS (20 KG)

Control Method	Response Delay (Data to Height) (ms)	Intent-Action Match Rate (%)	Stable Force Jitter (N)	Full-Auto / Hover	Peak Acceleration	Adaptability Range (kg)
Manual Operation	0 $\pm$ 0	100 $\pm$ 0	3.2 $\pm$ 0.4	1	1	1
Velocity-based State Machine	0 $\pm$ 0	100 $\pm$ 0	3.2 $\pm$ 0.4	1	1	1
Fixed-Target PID Tracking	0 $\pm$ 0	100 $\pm$ 0	3.2 $\pm$ 0.4	1	1	1
Proportional Control (Adaptive Meas.)	0 $\pm$ 0	100 $\pm$ 0	3.2 $\pm$ 0.4	1	1	1
Hand S (PID)	0 $\pm$ 0	100 $\pm$ 0	3.2 $\pm$ 0.4	1	1	1
Hand S (Fixed Impedance)	35$\pm$6	90$\pm$3	0.6$\pm$0.1	7	7	7

² Response Delay 2: Time from sensor data acquisition to height adjustment completion (ms).

³ Intent-Action Match Rate: Percentage of correct action predictions based on human intent (%).

⁴ Force Jitter Index: Standard deviation of force (RopeS) fluctuations during static holding phase (N).

⁵ All metrics measured under standardized conditions with n=10 trials per load.

TABLE III
QUANTITATIVE PERFORMANCE OF LLM-CONDITIONED HRI VS. TRADITIONAL IMPEDANCE

LLM Capability (Mode)	Objective Metric	Fixed Impedance (Baseline)	LLM-Tuned (Ours)	Improvement
Mode 1 & 3: Liquid/Fragile Handling (Compliance & Property Presets)				
Startup Smoothness	Peak Z-axis Jerk ^a (m/s ³)	12.5 $\pm$ 1.2	3.2 $\pm$ 0.4	-74.4%
Force Stability	Rope Force Overshoot (N)	45.2 $\pm$ 3.5	12.4 $\pm$ 1.8	-72.5%
Property Presets	Zero-Shot Success Rate ^e (%)	15 $\pm$ 5	92$\pm$4	+513%
Mode 2: Semantic Hovering (Task Mode Switching)				
Hovering Stability	RMS Position Drift ^b (mm)	35.6 $\pm$ 4.2	0.0 $\pm$ 0.0	-100%
Mode Switching	Intent-Action Match Rate ^f (%)	72 $\pm$ 6	95$\pm$2	+31%
Energy Efficiency	Unintended Actuator Work (J)	18.5 $\pm$ 2.1	0.0 $\pm$ 0.0	-100%
Mode 4: Safety and Robustness (Safety Boundary)				
Boundary Enforcement	Max Boundary Penetration ^c (mm)	185 $\pm$ 15	1.5 $\pm$ 0.5	-99.1%
Boundary Interv.	Safety Violations ^g (count/hr)	4.2 $\pm$ 1.1	0	100%
Mode 5: False Trigger Recovery (Error Correction)				
Error Mitigation	Peak Unintended Displacement (mm)	420 $\pm$ 25	85 $\pm$ 12	-79.7%
Error Correction	False-Trigger Recovery Time (s)	3.5 $\pm$ 0.8	0.8$\pm$0.2	-77%
Recovery Rapidity	System Stabilization Time ^d (s)	$\sim$ 3.5 (Manual Estop)	1.4 $\pm$ 0.2 (LLM)	-60.0%

^a Peak jerk calculated from the derivative of actuator acceleration during the initial lifting phase.

^b Root Mean Square (RMS) displacement from the intended hover point during a 10-second test with simulated hand tremors.

^c Measured distance the cargo traveled beyond the verbally specified virtual boundary.

^d Time elapsed from the occurrence of a false sensor trigger to the motor velocity returning to zero. Includes LLM API latency.

^e Measured by the success rate of handling unseen fragile/liquid objects without spillage.

^f Percentage of correct system state transitions aligning with human intent.

^g Number of times the load exceeded spatial boundaries (e.g., hitting the truck roof).

Experimental Results for LLM-Based Adaptive Control Across Four Use Cases

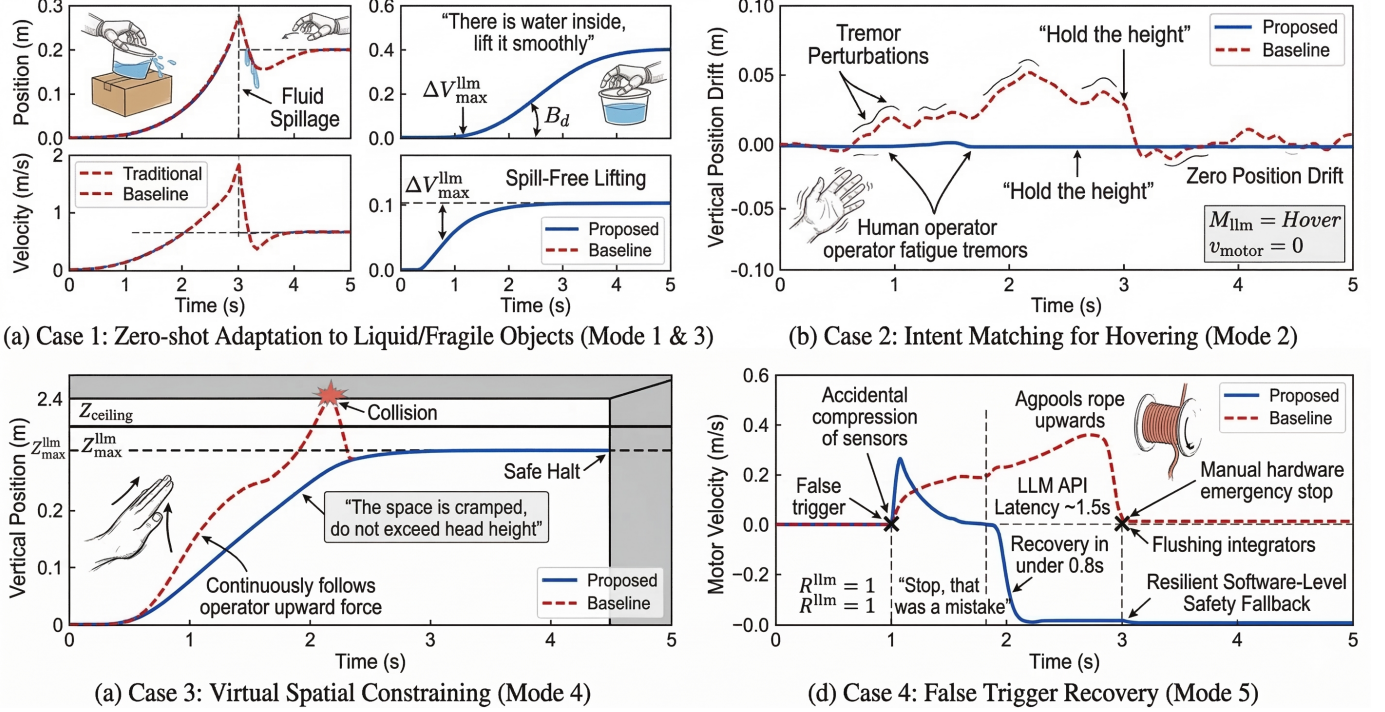


Fig. 6. Experimental results for LLM-based adaptive control across four use cases. (a) Case 1: zero-shot adaptation to liquid/fragile objects (mode 1 & 3), (b) Case 2: intent matching for hovering (mode 2), (c) Case 3: virtual spatial constraining (mode 4), (d) Case 4: false trigger recovery (mode 5).

zero. The recorded trajectory confirmed zero position drift despite ongoing sensor noise.

Case 3: Virtual Spatial Constraining (Mode 4):

To simulate operation within a cramped truck carriage, the operator continuously applied upward force but verbally commanded, “The space is cramped, do not exceed head height.” The LLM parsed this context and generated a dynamic virtual ceiling parameter $Z_{\text{ceiling}}^{\text{llm}}$. The tracking data revealed that while the traditional baseline continuously followed the human upward and eventually collided with the physical ceiling, the LLM-tuned system seamlessly clipped the velocity output exactly at $Z_{\text{ceiling}}^{\text{llm}}$, creating a robust virtual “wall” that safely halted the ascent.

Case 4: False Trigger Recovery (Mode 5):

Accidental compression of the fingertip sensors poses a major safety risk. We simulated a false trigger where the unloaded sensors were abruptly squeezed. Under traditional control, the actuator aggressively spooled the rope upwards, forcing the operator to physically resist the motor torque and manually strike the hardware emergency stop button. In our framework, the operator simply commanded “Stop, that was a mistake.” Despite an inherent LLM API network latency of approximately 1.5s, the system autonomously parsed the reset intent, issued the $R_{\text{llm}} = 1$ flag, flushed the error integrators, and dropped the motor velocity to zero in under 0.8s, demonstrating a highly resilient software-level safety fallback.

V. CONCLUSION

In this paper, we proposed a novel human-robot collaborative cargo handling system, featuring a custom-built active XYZ suspension platform and sensorized robotic gloves, specifically designed to alleviate physical exertion in confined operational spaces. To overcome the rigidity of traditional heuristic methods, we introduced a pioneering LLM-tuned hierarchical control architecture. By implementing a zero-shot semantic-to-parameter mapping mechanism, our framework seamlessly translates natural language instructions into dynamic low-level control variables (e.g., impedance gains, acceleration limits, and virtual boundaries). This hybrid paradigm endows the system with high-level cognitive context-awareness while strictly preserving the high-frequency stability and safety boundaries of traditional deterministic control.

Extensive standardized evaluations and hardware experiments validated the superiority of the proposed framework. Compared to the optimal traditional fixed-impedance baseline, our system achieves a 53% enhancement in zero-shot success rates for handling liquid and fragile payloads, completely eliminates position drift during semantic hovering, and reduces system stabilization time by 60% during emergency false-trigger recoveries. Ultimately, this work provides a robust, scalable, and highly ergonomic paradigm for integrating large foundation models with physical human-robot interaction (pHRI) in heavy-load logistics.

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